UTS Cloud Computing Class - Lab Guide

Week 6 Lab: Resources

This week and next week the lab tasks will involve configuring and setting up different Resource and permission mechanisms in a vSphere environment. These features are important in order to create a viable multi-tenancy solution that may involve different tenants receiving different permissions, and resource allocations

Task 1. Prepare for the lab.

1. Open VMware Workstation, and **power on** the following five/six VMs: 'ESXi01', 'ESXi02', 'VCSA' + 'OpenWRT' and 'iSCSI' ('DT' for F2F students).
2. Verify that all your machines are online and connected to the network.
3. Connect to your VCSA vm using a **web browser** (from 'DT') (https://172.20.10.94).
4. Power up the VMs that you have created using OVF on your ESXi hosts.

*** STOP *** *Ping from the Terminal (or cmd) window to check if there is a connection to ESXi01 and ESXi02. If connectivity is not established follow the steps outlined from earlier weeks before continuing.*

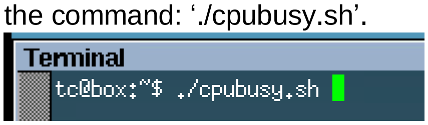
Task 2. Create a Resource Pool.

We will create two pools to show how a VM will receive different resources in different pools.

1. On the VMWare Web Client (VCSA), Select any Host with two VMs on it.
2. Create a new **Resource Pool** called "Fast" and assign it **High share** values.
   (Right click on the host IP -> New Resource Pool)
3. Create a new **Resource Pool** called "Slow", and assign it Low share values, and a **500Mhz Limit**.
4. Add one VM to the High Pool, and one VM to the Low Pool (make sure both VMs are on same host)
   (Simply drag and drop the VMs on the respective Resource Pool)

Task 3. Create CPU Workload on the VMs

Observe changes to the VM performance. This task is for you to observe how VMs consume physical device resources.
Currently you should have 2 VMs both running on 1 host, in different resource pools.

1. Open Terminal window in the virtual machine and run the cpubusy.sh script. On the **'$'** prompt type the command: './cpubusy.sh'.
   

2. How long does it take to process the script inside each VM? ______________________________
3. **Remove the Limit** on the Low pool. ______________________________
4. How long does it take to process the script inside each VM? _____________________________
5. Are the times different for the VMs in different pools? ______________________________
6. **Assign a Limit** to the **Fast** pool of 500 MHz. ___________________________________

7. How long does it take to process the script inside each VM? ________________________
8. Which VM is Faster at the moment? ______________________________
9. **Remove the limit** so that neither pool has a CPU limit.
10. How long does it take to process the script inside each VM? ________________________

Task 4. Force Competition.

To make the VMs compete, we must force them to use the same CPU. ESXi actively avoids doing this, but it is necessary for the lab demonstration.

1. Power Off both VMs.
2. On VMWare Web client (VCSA), **Edit Hardware settings** on each VM, and find the **Scheduling Affinity** the **CPU** section.
   (Right-click on vm -> Edit Settings -> Collapse CPU section -> Scheduling Affinity)
3. Type **0** into the field to instruct the system to run this VM on CPU0 only. Ensure this setting is configured for both VMs.
4. For Resource Pool, change the Share value to '**Normal**' for both the 'Fast' and 'Slow' resource pools.
5. **Power up** the VMs again.
6. Re-run the **cpubusy** script on both VMs.
7. How long does it take to process the script inside each VM?
8. Are the times different? Why?

Task 5. Create a Cluster to Prepare for the next lab.

Create a Cluster object in your datacenter and add your hosts.

1. Select your Datacenter, and right-click to create a Cluster called ***YourName*-Cluster**
2. **Drag** both your hosts into the Cluster.
3. Notice the VMs now appear under the cluster directly, not the hosts.
4. **Power Off** your VMs.
5. Remove the **Affinity** settings on both VMs. (make sure there is no values at all)
6. Shutdown the Hosts, and all your other External VMs.